Kaustubh Gupta

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RESEARCH INTERESTS

Areas: AI safety, multilingual LLM safety, agency-preserving alignment

Methods: mechanistic interpretability, activation steering, large-scale behavioural evaluation

EDUCATION

BTech in Artificial Intelligence and Engineering Science - 9.34 Hyderabad, IN | 2025

Indian Institute of Technology Hyderabad

Teaching Assistant: Convex Optimization (Jan-May 2024, Jan-May 2025)

Standardized Tests: IELTS Academic: Overall 8.0 (Listening 8.5, Reading 9.0, Writing 7.0, Speaking 7.0), CEFR C1 [Oct 2026]

Coursework: Probability, Convex Optimization, ML, Deep Learning, RL, NLP, Speech Systems, AI For Humanity

EXPERIENCE & FELLOWSHIPS

ORACLE | Data Scientist 1 Bangalore, IN | Jul 2025 – Present

- Ship LLM query-understanding for Fusion Analytics Warehouse Copilot: entity classification, column reranking, and retrieval-backed generation.
- Improve Copilot quality via prompt tuning, JSON response contracts, and production backend-log analysis.

SENTIENT FUTURES | Research Fellow Remote | Aug 2026 – Present

- Lead Workstream B (mentors: John Lund, AE Studio, and CaML) on whether steering along persona vectors can protect moral consideration in LLMs under adversarial fine-tuning. Built the steering and LoRA-erosion pipeline on Qwen3-8B, where steering raises the trait score from 63.8 to 87.5 with coherence intact.

BETTER FUTURES FELLOWSHIP | Research Fellow Remote | Aug 2026 – Present

- Studied delegation drift, where models take over decisions from users across multi-turn conversations. Evaluated 5 open-weight models over 720 trajectories (2,880 turns) and found engagement-oriented prompts cause a +23–25% decision-takeover cliff versus autonomy prompts.

ORACLE | Data Analytics Intern Bangalore, IN | May 2024 – Jul 2024

- Built LLM synthetic-data pipeline for Oracle FAW using Python and K-Means/farthest-point sampling, adopted by a Consulting MTS.

HONORS & AWARDS

- Top 100, Meta × PyTorch × Hugging Face OpenEnv Hackathon (31,000+ teams), Apr 2026.
- Highest CGPA in Department (9.34), BTech AI, IIT Hyderabad, 2025.

PUBLICATIONS

Interpreting and Steering State-Space Models via Activation Subspace Bottlenecks | ICML 2026

Second author. Mechanistic interpretability on Mamba LLMs via activation subspace bottlenecks, benchmarking steering controllability vs. quality.

Stream-TTS: A Low-Latency Text-to-Speech using Kolmogorov-Arnold Networks | IEEE ICASSP 2025

Co-authored multilingual streaming TTS with KANs, reaching 0.0795 RTF and 4.09 MOS with prosody control.

Fairness-Aware Submodular Optimization: A Survey | Unpublished survey, IIT Hyderabad

Sole-authored survey of fairness constraints in submodular maximization (summarization, influence spread). Synthesized fairness–utility tradeoffs and open directions for equitable ML selection.

Attentive Multimodal Learning for Suicide Risk Assessment Using Acoustic and Prosodic Cues | Under review, Speech Communication

Built initial multimodal pipeline with acoustic embeddings and prosodic cues for suicide risk classification.

PROJECTS

MULTI-TASK MULTILINGUAL LLM REFUSAL EVALUATION

Python, PyTorch, HuggingFace, OR-Bench

With Prof. Usman Naseem (Macquarie Univ.). Tests whether task wrappers shift LLM refusal across languages. Built a 211k-row benchmark (1,761 seeds × 15 languages × 8 task wrappers) extending Sakonii task-over-refusal, replicated the baseline on Llama-3.1-8B with a calibrated OR-Bench-style judge, and now evaluating task-conditioned refusal shifts.

PREFKIT: MULTI-METHOD PREFERENCE ELICITATION TOOLKIT

Python, PyTorch, HuggingFace

Tests whether stated model preferences depend on how they are elicited. Built a toolkit with 4 elicitation methods (pairwise, Likert, action-choice, best-worst scaling) and a Cross-Method Score, and ran it across Qwen3 (1.7B–14B) on a 24-outcome set for the Apart Digital Minds Sprint (Track 4).

DETECTION IS NOT CONTAINMENT: TIMELY MONITORING OF AI AGENT FAILURES

Python, JSONL, Incident Forensics, Response Protocol

Apart Research × CeSIA AI Incident Response Sprint (Track 2). Asks whether monitoring can stop an agent failure in time. Co-built a public-source incident claim-extraction pipeline and a monitoring-intervention experiment showing that detecting an agent action only helps if a last-safe point to stop it remains, including the cost of false positives.

OPTICAL CHARACTER RECOGNITION FOR ANCIENT SANSKRIT MANUSCRIPTS

Python, OCR, CRNN, Multimodal LLMs

Digitizes degraded Sanskrit manuscripts. Built an OCR pipeline with line segmentation and CRNN fine-tuning, and compared it against multimodal LLMs.

AI-POWERED SESSION HIJACKING PREVENTION

Python, XGBoost, Flask, Redis

Detects hijacked web sessions from HTTP traffic. Built an XGBoost anomaly detector on CSIC 2010 (F1 0.97 with SMOTE), served through Redis-backed Flask inference with an admin dashboard.

SKILLS

Languages: Python, C++, C, SQL, JavaScript, Bash

ML & Research: PyTorch, HuggingFace, TransformerLens, activation steering, linear probing, LoRA fine-tuning, LLM evaluation (LLM-as-judge)

Tools: Git, Docker, Ollama, Flask, Redis, XGBoost, LaTeX